Transportation Research Part D 126 (2024) 104001

Contents lists available at ScienceDirect

Transportation Research Part D

journal homepage: www.elsevier.com/locate/trd

Commuting and its spillover effects on subjective well-being: Evidence from China

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ARTICLE INFO

Keywords:
Subjective well-being
Commuting mode
Life domains
Structural equation model
China

ABSTRACT

The relationship between commuting and subjective well-being has long been studied in the transportation field. However, few studies have examined the complex relationship between commuting characteristics that affect subjective well-being, nor have they focused on the spillover effects of commuting on other life domains. To fill these gaps, we built a structural equation model using large-scale survey data across 29 provinces in China to explore the mechanisms by which commuting characteristics affect subjective well-being. The results suggest that the complex relationship between commuting mode, time, and cost reduces the differences in commuters' subjective well-being across different commuting modes. Commuting distance affects commuting choices and indirectly produces different subjective well-being outcomes. The impact of commuting spills over into the health, work, and community domains, affecting subjective well-being. Based on these findings, we provided policy recommendations for urban and transportation planning in Chinese cities.

1. Introduction

Subjective well-being (SWB) refers to people's perception of their quality of life and includes life satisfaction and affective components (Diener, 1984; Veenhoven, 2015; Liu et al., 2017; Zhu et al., 2019). As a valid proxy for personal welfare (Di Tella and MacCulloch, 2006; Kahneman and Krueger, 2006; Crespo and Mesurado, 2015), subjective well-being has attracted interest from researchers across multiple fields (Zijlstra and Verhetsel, 2021). Commuting is a daily routine that impacts people's lives (Lorenz, 2018; Zhu et al., 2019). It has been identified as an important determinant of both life satisfaction (Wang et al., 2021b) and subjective well-being (Nie and Sousa-Poza, 2018). Fully understanding the relationship between commuting and subjective well-being can guide policy development and planning management to improve people's subjective well-being.

Numerous research in recent years has provided evidence for how commuting characteristics affect subjective well-being (St-Louis et al., 2014; Yin et al., 2019; Zhu et al., 2019; Herman and Larouche, 2021; Zijlstra and Verhetsel, 2021; Sun et al., 2021). However, these studies have some limitations. Most studies used regression models to evaluate the effects of single commuting characteristics on subjective well-being, neglecting the possible mediating effects between commuting distance, mode, time, and cost. The impact of commuting on subjective well-being through its influence on other life domains is termed the spillover effect of commuting (Chatterjee et al., 2020; Tao et al., 2022). VanderWeele (2017) divided daily life into five main domains: spirituality, family, work, health, and community. However, research on how the effects of commuting spill over into these specific life domains (e.g., health, work, and

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https://doi.org/10.1016/j.trd.2023.104001
Received 9 June 2023; Received in revised form 15 August 2023; Accepted 29 November 2023
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community) remain limited. Moreover, the data used in these studies are mostly from developed Western countries. Although research in China has been growing in recent years, most studies have focused on specific cities and may not reflect the influence mechanism of the entire country (Zhu and Fan, 2018; Sun et al., 2022).

In this study, we developed a structural equation model (SEM) to analyze the relationship between commuting and subjective well-being. The contributions of this study are as follows: (1) We investigated the mediating effects of commuting mode, cost, and time on subjective well-being; (2) We explored how commuting affects subjective well-being by spilling over to other life domains, including health, work, and community; (3) We used a dataset that covers entire China to provide more representative results.

The remainder of the paper is structured as follows: Section 2 provides a literature review to establish the background for the study; Section 3 introduces the data and methodology; Section 4 presents the analysis of the results; Section 5 discusses the empirical findings; and the last section draws conclusions and provides policy recommendations.

2. Literature review

This section investigates the findings of previous studies on the relationship between commuting characteristics and subjective well-being in terms of empirical results, spillover effects, and research data.

Some studies have examined the relationship between commuting characteristics and subjective well-being. The majority of these studies show that active commuters tend to report higher subjective well-being compared to commuters who use motorized modes, such as cars, metros, and buses (Mao et al., 2016; Ye and Titheridge, 2017; Zhu and Fan, 2018; Handy and Thigpen, 2019; Mouratidis et al., 2019; Ye et al., 2020). However, a study by Zhu et al. (2019) presented contradictory findings. They found that residents who commute by motorized modes in rural areas of China report higher subjective well-being, while no significant differences were observed in urban areas. These findings suggest that the effect of commuting mode on subjective well-being may be influenced by an individual's residential location. The impact of commuting time on subjective well-being has also been of interest. Several studies have demonstrated that commuting time has a major negative impact on subjective well-being and reducing commuting time is an effective approach to improving subjective well-being (Kahneman et al., 2004; Stutzer and Frey, 2008; Roberts et al., 2011; Choi et al., 2013; Hilbrecht, 2014; Wheatley, 2014). However, some recent studies have concluded that commuting time is not significantly negatively related to residents' subjective well-being (Dickerson et al., 2014; Morris and Zhou, 2018; Clark et al., 2020). This is because a longer commute may result in certain compensatory that offset the travel time loss, such as higher wages, more desirable job opportunities, lower housing prices, better housing quality, or shorter commuting times for family members (Green, 1997; So et al., 2001; Plaut, 2006; Sandow, 2008; Sandow and Westin, 2010). In terms of research methods, most of the existing studies have used regression models such as ordered regression models and multilevel regression models. Most of these studies only examine the effects of commuting mode on subjective well-being, and overlook the contribution of commuting time in the effects of commuting mode on subjective well-being. The findings of Lades et al. (2020) showed that the higher subjective well-being of active modes was mostly caused by the short duration. Once the difference in duration was controlled, the effect of active modes on subjective well-being would be diminished. However, few researchers have paid attention to the complex relationship of commuting characteristics on subjective well-being.

Research has explored the connections between travel behavior and various aspects of life (Tao et al., 2022). Commuting behavior not only affects satisfaction with commuting (Chatterjee et al., 2020), but also influences activities and satisfaction in other life domains, with long-term effects on subjective well-being (De Vos et al., 2013). This effect is referred to as the spillover effect of commuting on other life domains (De Vos et al., 2013; Chatterjee et al., 2020; Tao et al., 2022; Ye et al., 2022). On the one hand, spillover effects are manifested in the impact of longer commuting times on other activities. For example, studies by Hilbrecht (2014) and Nie and Sousa-Poza (2018) implied that longer commuting times reduce the time available for sports or other daily activities, resulting in lower subjective well-being. Urry's research indicated that social participation indirectly contributed to the impact of travel on well-being (Urry, 2002). On the other hand, it also manifests itself in changes in mood after commuting and effects on domain satisfaction. For example, Delmelle et al. (2013) showed that longer commuting times were associated with lower social satisfaction. VanderWeele (2017) identified major life domains including spirituality, family, work, health, and community. Satisfaction with these life domains has been found to affect subjective well-being (Gao et al., 2022), representing a bottom-up spillover effect (Sirgy, 2001). However, there is little existing literature investigating how the effects of commuting spill over into specific life domains and subsequently affect subjective well-being.

Research on the relationship between commuting characteristics and subjective well-being has primarily relied on data from developed European and American nations. However, it is crucial to verify whether these conclusions can be applied to China, given the differences in development backgrounds and stages. In recent years, there has been a growing body of literature on commuting and subjective well-being in Chinese cities, shedding light on the characteristics and effects of commuting within this context. Several studies devoted to exploring the structural relationship among commuting behavior, satisfaction, and subjective well-being, finding that commuting characteristics significantly affect commuting satisfaction and further influence subjective well-being (Ye and Titheridge, 2020; Sun et al., 2022). Some studies have identified potential mediators in the relationship between commuting characteristics, satisfaction, and subjective well-being, such as commuters' personalities, moods, and other domains of satisfaction (Gao et al., 2017a, 2017b). Overall, these studies provide additional perspectives on the relationship between commuting and subjective well-being. Nevertheless, most of these studies have used data from a single city, such as Shanghai (Sun et al., 2022) and Xi'an (Zhu and Fan, 2018). The results of these studies only reflect the circumstances of a specific city and do not provide a comprehensive understanding of the overall influence mechanism in Chinese cities. Furthermore, they cannot provide policy inspiration for other cities. While there are studies that have employed datasets covering the entire China, most of them have focused on a single commuting

feature, such as the effect of commuting time on subjective well-being (Ma et al., 2018; Zhu et al., 2019) or the indirect effects of commuting time on subjective well-being through other life domains (Sun et al., 2021). Consequently, using nationally representative data and constructing research models that consider multiple commuting characteristic variables can provide more generalizable results and help verify the consistency of previous findings.

3. Data and methodology

3.1. Conceptual model

In light of existing literature and common knowledge, we constructed a conceptual model to investigate the relationship between commuting and subjective well-being. As shown in Fig. 1, three sorts of elements that may affect subjective well-being are included in the conceptual model.

The first category is socio-demographic attributes, including individual and household characteristics (i.e., gender, age, education, personal and family income, hukou, marriage, car ownership, household size, and presence of elderly and children) and housing characteristics (i.e., house location and house type). We incorporated these socio-demographic characteristics directly into the model as control variables as suggested by the literature, given their significant direct impact on subjective well-being (Archer et al., 2013; Clark et al., 2019). It is important to note that these socio-demographic characteristics may also have an impact on commuting mode choices (X. Wang et al., 2020; Wang et al., 2021a). To account for potential variations among different counties, we added average annual PM2.5, average annual temperature, and standard deviation of temperatures as control variables (Yang et al., 2021).

The second category is commuting characteristics, including commuting time, commuting cost, commuting distance, and commuting mode (i.e., walking, bike, e-bike, bus, metro, and car). Commuting mode, cost, and time directly affect subjective well-being (Yin et al., 2019; Zhu et al., 2019; L. Wang et al., 2020; Zijlstra and Verhetsel, 2021; Sun et al., 2021; Wang et al., 2021b). The complex relationship between commuting characteristics that affect subjective well-being follows a decision-making process with a long-medium-short time horizon (Deng and Zhao, 2022). Individuals first choose their residences and workplaces, which determines the commuting distance. Subsequently, they choose the commuting mode accordingly, which determines the associated commuting cost and time. Commuting distance can also directly affect commuting cost and time. In other words, commuting mode indirectly affects subjective well-being through commuting cost and time (Sun et al., 2022). Commuting distance indirectly affects subjective well-being through commuting mode, cost, and time (Sun et al., 2022; Deng et al., 2023).

The third category is life domains, including self-reported health, working hours, relationship spending, relationship income, and social trust. Commuting time can affect subjective well-being by influencing other life domains (Chatterjee et al., 2020). Thus, we included relevant variables that characterize the life domains of health, work, and community as mediators to explore how commuting affects subjective well-being through these life domains. It should be noted that although we employed a causal relationship direction from commuting time to these life domains, reverse causality is also possible. For example, self-reported health and working hours might impact commuting time rather than the reverse. However, with cross-sectional data alone, we cannot establish a definitive causal relationship. Since most of the literature adopted the causal relationship direction from commuting to other life domains

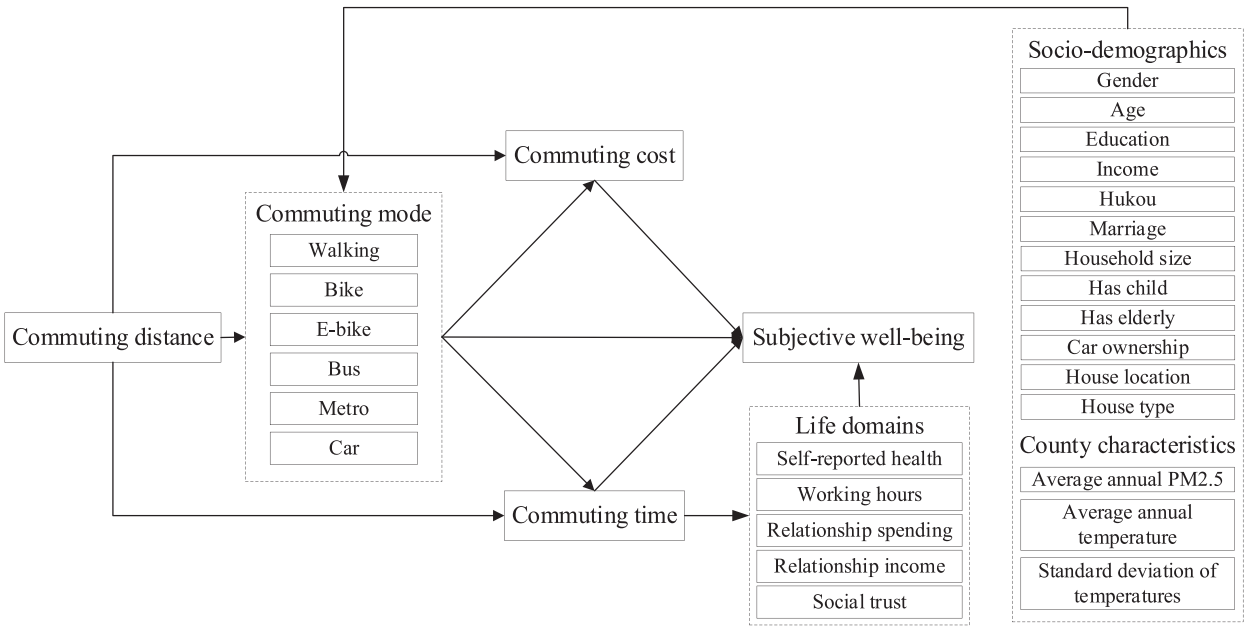

Fig. 1. Conceptual model.

(Schwanen and Mokhtarian, 2005; Heinen et al., 2013; Mattisson et al., 2018), we adopted the same causal relationship direction whereby commuting time affects subjective well-being by influencing the health, work, and community domains.

3.2. Data

The data for this study were obtained from the China Household Finance Survey (CHFS) conducted by the Southwest University of Finance and Economics in China. It is a nationally representative survey, employing a stratified, three-stage probability proportional to size (PPS) sampling technique, covering 29 provinces. The CHFS has been an ongoing initiative since 2011, and for our analysis, we used data from the 2017 CHFS, which encompassed 40,011 families in 355 counties. The CHFS utilizes a standardized questionnaire administered through a face-to-face interview with the individual in each household who possesses the most knowledge about their financial status such as housing assets, financial wealth, debt and credit constraints, income and consumption, social security and insurance, intergenerational transfers, demographic characteristics, employment, and payment habits. These data contain information regarding demographic characteristics, commuting characteristics, housing circumstances, and subjective attitudes, making them suitable for our study.

The CHFS dataset was initially used in studies of Chinese household finance, assets, debt, and consumption. Over time, its usage has extended to studies of subjective well-being, such as the association with housing wealth, homeownership, digital finance, and mobile payments (Zhang and Zhang, 2019; Zheng et al., 2020). The CHFS dataset has only recently been used in transportation research, such as the influencing factors of car ownership, the determinants of shared bike use, and the impact of residential self-selection on commuting modes (Le Vine et al., 2018; Deng and Zhao, 2022, 2023). Nonetheless, there appears to be no research linking transportation to subjective well-being using the CHFS dataset.

We also incorporated atmospheric environmental and temperature data to account for potential variations among different counties. Atmospheric environmental data were obtained from the national urban air quality real-time dissemination platform of the China general environmental testing station. Temperature data were derived from the fifth-generation atmospheric reanalysis dataset (ERA5) of the European Center for Medium-Range Weather Forecasts (ECMWF).

3.3. Variables

Table 1 displays the variables and the descriptive statistics. To control for respondents' personal and household characteristics as well as housing characteristics, 13 socio-demographic variables (i.e., gender, age, education, hukou, marriage, personal and family income, car ownership, household size, presence of elderly and children, house location, and house type) are used. Gender is a nominal variable with two options (female and male). Age is a quantitative variable directly reported by the respondents. Education represents the highest level of education of the respondents, which is divided into nine categories in the questionnaire, ranging from never attend school to doctorate. We processed it into an ordered categorical variable from 1 to 7 based on the common schooling stages in China, representing never attend school, primary school, secondary school, high school/polytechnic, junior college/university undergraduate, master's, and doctorate, respectively. Hukou indicates the household registration status, which is also a nominal variable with two options (local and non-local hukou). Marriage is also a nominal variable with two options (single and married). Personal and family income are quantitative variables measured on an annual basis, which are directly reported by the respondents. Car ownership is the number of cars owned by the household, directly reported by the respondents. Household size is the number of family members directly reported by the respondents. The presence of elderly and children are nominal variables with two options (yes and no), providing some indication of household structure. House location is divided into five categories in the questionnaire (i.e., urban areas of large cities, suburban areas of large cities, large towns, small towns, and rural), which are completed by visitors who observed or asked people in the community or village. For our analysis, we consolidated them into three categories (large city, town, and rural). Large cities and towns are represented by separate nominal variables, with rural serving as the reference level. Lastly, housing type is a nominal variable with two options (owned and rented).

There are four variables in commuting characteristics (i.e., commuting distance, mode, cost, and time). The questionnaire provides ten consecutive options from less than 2 km to more than 60 km for commuting distance. To quantify this variable, we assigned the maximum value of each range to the corresponding segments. For example, less than 2 km was assigned 2 km, 2–5 km was assigned 5 km, and more than 60 km was assigned 80 km. The questionnaire provides nine consecutive options for commuting time, from less than 15 min to more than 120 min. We assigned the maximum value of each range to the corresponding segments. For example, less than 15 min was assigned 15 min, 15–30 min was assigned 30 min, and more than 120 min was assigned 150 min. In terms of commuting cost, the questionnaire provides seven consecutive options ranging from less than 5 Yuan to more than 100 Yuan, along with a no-cost option. Similarly, we assigned the maximum values of each range to the corresponding segments. For example, no-cost was assigned 0, less than 5 was assigned 5, 5–10 was assigned 10, and more than 100 was assigned 150. Regarding commuting mode, the questionnaire provides eight commuting modes (i.e., bus, rail, shuttle bus, private car, taxi, e-bike, bike, and walking). Since the share of shuttle buses and taxis is relatively small, we grouped them with buses and private cars, respectively. As a result, we obtained six commuting modes (i.e., bus, rail, car, e-bike, bike, and walking), with the car mode serving as the reference level.

To characterize residents' health, work, and community domains, we used several variables, including self-reported health, working hours, relationship spending, relationship income, and social trust. Self-reported health indicates the respondent's perception of their physical condition compared to their peers. It was measured using a five-point Likert scale (1 = very bad, 2 = bad, 3 = fair, 4 = good, 5 = very good). Working hours are quantitative variables that are directly reported by the respondents. The community domain considers personal relationship maintenance and social trust, which describe the respondents' social network relations from the

perspective of acquaintances and strangers, respectively. Relationship maintenance includes relationship spending and relationship income. Relationship spending is the monetary amount given to non-family members over the course of the previous year due to holidays, weddings and funerals, education, living expenses, donations or grants, medical care, and other matters. Relationship income is the monetary amount received from non-family members due to the same reasons, except that medical care becomes inheritance. Social trust is a key factor in the relationships between individuals and groups (Bai et al., 2019). The survey asked respondents to indicate their level of trust in strangers using a five-point Likert scale (1 = very distrustful, 2 = distrustful, 3 = fair, 4 = trustful, 5 = very trustful). These variables were included in the model as mediating variables to estimate the effect of commuting time on subjective well-being.

Subjective well-being was measured by a single subjective attitude question: "In general, do you feel happy now?" Respondents rated their happiness on a five-point Likert scale. To ensure consistency in the order of the scales within the model, we inverted the order of the scale (1 = very unhappy, 2 = unhappy, 3 = average, 4 = happy, 5 = very happy).

3.4. Methodology

To reveal the complex relationship between commuting and subjective well-being, we used the structural equation model (SEM). The structural equation model is a widely-used modeling tool for analyzing intricate multivariate relationships and has gained popularity in both travel behavior and subjective well-being studies (Friman et al., 2017; Ye and Titheridge, 2017; Yin et al., 2021). It

Table 1
Descriptive statistics of variables.

Variable	Mean/Cases	Std.Dev./Percentage
Age	43.5	1.16
Gender		
Male	5611	54.76%
Female (Ref)	4636	45.24%
Education	4.57	1.88
Hukou		
Local	8914	87.00%
Non-local (Ref)	1333	13.00%
Family income, ten thousand Yuan	8.5	0.98
Personal income, ten thousand Yuan	4.6	0.58
Marriage		
Married	8521	83.31%
Single (Ref)	1707	16.69%
Household size	3.15	1.30
Has children		
Yes	3444	33.61%
No (Ref)	6803	66.39%
Has elderly		
Yes	7935	77.44%
No (Ref)	2312	22.56%
Car ownership	0.46	0.60
House location		
Large city	6474	63.18%
Town	1362	13.30%
Rural (Ref)	2411	23.52%
House type		
Owned	8607	84.04%
Rented (Ref)	1635	15.96%
Average annual PM2.5, μg/m3	43.82	14.60
Average annual temperature, ℃	15.34	5.20
Standard deviation of temperatures, ℃	8.79	2.63
Self-reported health	3.76	0.86
Working hours, hour	8.46	2.20
Relationship spending, ten thousand Yuan	0.4	0.09
Relationship income, ten thousand Yuan	0.3	0.15
Social trust	2.26	0.88
Commuting cost, yuan	6.4	1.35
Commuting time, min	0.49	0.33
Commuting distance, km	8.8	1.15
Commuting mode		
Walking	2724	26.58%
Bike	955	9.32%
E-bike	2866	27.97%
Bus	1900	18.54%
Metro	497	4.85%
Car (Ref)	1305	12.74%
Subject well-being	3.79	0.78

allows for the testing of theory-driven hypotheses, unraveling the complex relationships between endogenous and exogenous variables, and quantifying their direct, indirect, and total effects.

In our study, the structural equation model was estimated using covariance analysis methods instead of ordinary least squares (OLS) linear regression (Yin et al., 2021). Since the commuting mode, the endogenous variable in the model is categorical, we used diagonally weighted least squares estimation rather than the commonly used maximum likelihood. Diagonally weighted least squares provide more accurate parameter estimates for non-normal endogenous variables, as they do not assume a specific distribution for these variables (Mîndrilă, 2010; Deng and Zhao, 2022). We implemented the diagonally weighted least squares estimation of the structural equation model using the lavaan package in R.

4. Results

The structural equation model in this study contains only structural models, not measurement models. The root mean square error of approximation (RMSEA) is 0.036, less than 0.08, indicating a good fit. The comparative fit index (CFI) is 0.890 and the Tucker-Lewis index (TLI) is 0.948, indicating a good fit.

Table 2 presents the standardized coefficients (without parentheses) and unstandardized coefficients (in parentheses) for each path in the model. The unstandardized coefficient β indicates that a one-unit change in the explanatory variable results in a β-unit change in the dependent variable, while the standardized coefficients β' reflect that a one standard deviation change in the explanatory variable results in a β' standard deviation change in the dependent variable. We used standardized coefficients to describe the effects of all variables, except dummy variables, because the standardized coefficients are suitable for comparing the effects of different explanatory variables on the dependent variable. For dummy variables such as commuting mode, we used unstandardized coefficients, because a one-unit change in a dummy variable means a shift from zero to one, while a one-standard-deviation change in a dummy variable is meaningless. Fig. 2 displays the path diagram between commuting characteristics and subjective well-being in the structural equation model.

4.1. Direct effects

Socio-demographic attributes and county characteristics have a significant effect on subjective well-being. Age, education, family income, and household size positively impact subjective well-being, while personal income negatively impacts subjective well-being. Women, homeowners, people without children or elderly living with them, and people living with partners exhibit higher subjective well-being, whereas people living in big cities exhibit lower subjective well-being. Average annual PM2.5 has a negative effect on subjective well-being, while the standard deviation of temperatures has a positive effect on subjective well-being.

Commuting characteristics are significantly related to subjective well-being. Commuting time and cost are negatively related to subjective well-being, which indicates that longer commuting times and higher costs reduce respondents' subjective well-being. Using commuting modes other than cars has a negative impact on subjective well-being. This suggests that people who commute by car have the highest subjective well-being. Among other commuting modes, people riding e-bikes and walking for commuting have higher subjective well-being, followed by metro, bus, and bike users.

Life domain variables all have a significant effect on subjective well-being. Consistent with previous research, self-reported health is positively related to subjective well-being, while working hours are negatively related to subjective well-being. Relationship spending, relationship income, and social trust are all positively related to subjective well-being.

4.2. Indirect and total effects

Table 3 shows the indirect and total effects of commuting characteristics on subjective well-being. Commuting distance has a significant indirect effect on subjective well-being by affecting commuting mode choice, time, and cost. Commuting mode has a significant indirect effect on subjective well-being through commuting time and cost. Since the magnitude of its indirect effects is smaller than that of its direct effects, the total effects of various commuting modes on subjective well-being remain to be significantly negative. Commuting time has a significant indirect effect on subjective well-being by affecting self-reported health and working hours. In terms of community domain variables, although the indirect effect through social trust was insignificant, the indirect effects through relationship spending and relationship income are significant, with a negative indirect effect through relationship spending and a positive indirect effect through relationship income. These results confirm that the impact of commuting can spill over to other life domains such as health, work, and community and then affect subjective well-being.

5. Discussion

Many studies have confirmed that commuting mode has a direct impact on subjective well-being (Mao et al., 2016; Ye and Titheridge, 2017; Zhu and Fan, 2018; Zhu et al., 2019; Handy and Thigpen, 2019; Singleton, 2019; Zijlstra and Verhetsel, 2021). We found that commuting time and cost play a significant mediating role in the effect of commuting mode on subjective well-being and that the indirect effect has the opposite sign of the direct effect. Consequently, the difference between the total effects of other commuting modes and car on subjective well-being was smaller compared to the difference between their direct effects on subjective well-being. Car commuters still have the highest subjective well-being after considering indirect effects, and the order of the remaining

Table 2
The standardized and unstandardized parameters of the structure equation model.

Variable	Commuting mode					Commuting time	Commuting cost	Self-reported health	Working hours	Relationship spending	Relationship income	Social trust	Subjective well-being
	Walking	Bike	E-bike	Bus	Metro								
Age	0.111[a]	0.084[a]	−0.068[a]	−0.001	−0.034[a]								0.053[a]
	(0.043)	(0.021)	(−0.027)	(−0.000)	(−0.006)								(0.047)
Gender: male	0.059[a]	0.007	−0.069[a]	0.056[a]	0.024[b]								0.025[b]
(Ref: female)	(0.052)	(0.004)	(−0.062)	(0.043)	(0.010)								(0.052)
Education	0.014	−0.004	−0.162[a]	0.091[a]	0.125[a]								0.049[a]
	(0.003)	(−0.001)	(−0.039)	(0.019)	(0.014)								(0.027)
Hukou: local	0.052[a]	−0.003	−0.049[a]	−0.034[a]	0.042[a]								0.016
(Ref: non−local)	(0.068)	(−0.002)	(−0.065)	(−0.039)	(0.026)								(0.050)
Family income	−0.011	0.020	0.002	−0.011	0.061[a]								0.038[c]
	(−0.005)	(0.006)	(0.001)	(−0.004)	(0.013)								(0.040)
Personal income	−0.039[b]	−0.046[b]	−0.062[a]	−0.029[c]	0.049[a]								−0.036[c]
	(−0.029)	(−0.023)	(−0.048)	(−0.019)	(0.018)								(−0.063)
Marriage: married	−0.044[a]	−0.007	0.053[a]	−0.008	−0.032[a]								0.102[a]
(Ref: single)	(−0.052)	(−0.006)	(0.064)	(−0.008)	(−0.018)								(0.279)
Household size	−0.003	−0.003	0.029[c]	−0.000	−0.016								0.097[a]
	(−0.001)	(−0.001)	(0.010)	(−0.000)	(−0.003)								(0.076)
Car ownership	−0.052[a]	−0.049[a]	−0.079[a]	−0.079[a]	−0.069[a]								0.012
	(−0.038)	(−0.023)	(−0.059)	(−0.050)	(−0.024)								(0.021)
Has children: yes	−0.014	0.006	0.054[a]	−0.018	−0.061[a]								−0.116[a]
(Ref: no)	(−0.011)	(0.004)	(0.051)	(−0.015)	(−0.028)								(−0.252)
Has elderly: yes	−0.008	−0.001	0.005	0.007	−0.001								−0.063[a]
(Ref: no)	(−0.008)	(−0.001)	(0.005)	(0.006)	(−0.000)								(−0.155)
House location (Ref: rural)													
Big city	0.033[b]	0.086[a]	−0.220[a]	0.172[a]	0.049[c]								−0.028[c]
	(0.030)	(0.051)	(−0.204)	(0.136)	(0.022)								(−0.060)
Town	0.020[c]	0.034[b]	−0.069[a]	0.050[a]	−0.038								−0.009
	(0.026)	(0.029)	(−0.091)	(0.057)	(−0.024)								(−0.027)
House type:	−0.002	0.034[a]	−0.028[b]	0.002	0.017[c]								−0.064[a]
owned	(−0.002)	(0.027)	(−0.034)	(0.002)	(0.010)								(−0.180)
(Ref: rented)													
Average annual	−0.024[c]	0.092[a]	0.034[a]	−0.007	−0.010								−0.043[a]
PM2.5	(−0.001)	(0.002)	(0.001)	(−0.000)	(−0.000)								(−0.003)
Average annual	−0.211[a]	0.009	0.187[a]	−0.097[a]	0.085[a]								0.007
temperature	(−0.018)	(0.000)	(0.016)	(−0.007)	(0.003)								(0.001)
Standard	−0.163[a]	0.036	0.056[b]	0.003	0.064[b]								0.072[a]
deviation of temperatures	(−0.027)	(0.004)	(0.010)	(0.000)	(0.005)								(0.028)
Commuting	−0.314[a]	−0.111[a]	−0.050[a]	0.313[a]	0.146[a]	0.679[a]	0.255[a]						
distance	(−0.121)	(−0.028)	(−0.020)	(0.105)	(0.027)	(0.194)	(0.299)						
Commuting mode (Ref: car)													
Walk						0.091[a]	−0.751[a]						−0.187[a]
						(0.067)	(−2.285)						(−0.433)
Bike						0.107[a]	−0.499[a]						−0.151[a]
						(0.121)	(−2.313)						(−0.532)
E-bike						0.105[a]	−0.683[a]						−0.183[a]
						(0.077)	(−2.049)						(−0.416)

(*continued on next page*)

Table 2 (*continued*)

Variable	Commuting mode					Commuting time	Commuting cost	Self-reported health	Working hours	Relationship spending	Relationship income	Social trust	Subjective well-being
	Walking	Bike	E-bike	Bus	Metro								
Bus						0.266[a]	−0.566[a]						−0.188[a]
						(0.227)	(−1.983)						(−0.502)
Metro						0.211[a]	−0.231[a]						−0.095[a]
						(0.326)	(−1.466)						(−0.457)
Commuting time								−0.048[a]	0.030[a]	−0.025[b]	0.084[a]	−0.009	−0.024[c]
								(−0.121)	(0.198)	(−0.006)	(0.038)	(−0.024)	(−0.076)
Commuting cost													−0.061[a]
													(−0.047)
Self-reported health													0.213[a]
													(0.264)
Working hours													−0.049[a]
													(−0.023)
Relationship spending													0.048[a]
													(0.599)
Relationship income													0.071[a]
													(0.490)
Social trust													0.031[a]
													(0.038)

Standardized coefficients are presented without parentheses; unstandardized coefficients are presented in parentheses. a, b, and c denote significance at the 0.01, 0.05, and 0.1 levels, respectively.

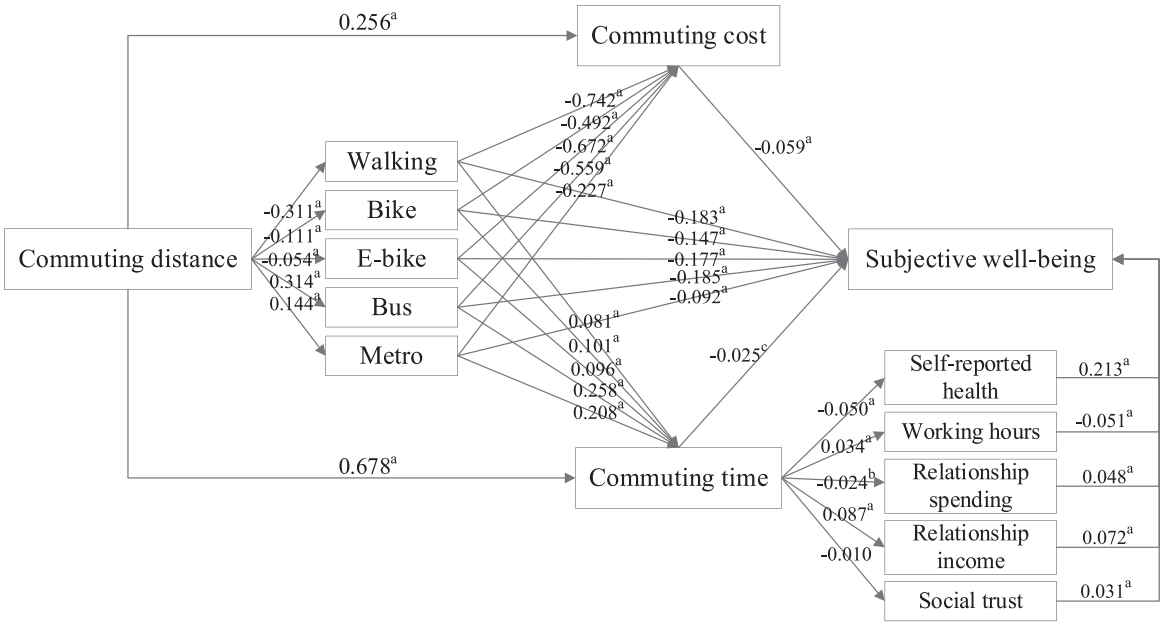

Fig. 2. The path diagram of the structure equation model. Note: a, b, and c denote significance at the 0.01, 0.05, and 0.1 levels, respectively.

Table 3
Indirect and total effects of commuting characteristics on subjective well−being.

Variable	Direct effect	Indirect effect (by mediating variables)	Total effect
Commuting distance		Walk: 0.040[a]; Bike: 0.012[a]; E-bike: 0.006[a]; Bus: −0.045[a]; Metro: −0.011[a]; Commuting time: −0.016[c]; Commuting cost:-0.016[a];	
Commuting mode			
Walk	−0.433[a]	Commuting time: −0.005[c]; Commuting cost: 0.106[a];	−0.332[a]
Bike	−0.532[a]	Commuting time: −0.009[c]; Commuting cost: 0.108[a];	−0.434[a]
E-bike	−0.416[a]	Commuting time: −0.006[c]; Commuting cost: 0.095[a];	−0.327[a]
Bus	−0.502[a]	Commuting time: −0.017[c]; Commuting cost: 0.092[a];	−0.427[a]
Metro	−0.457[a]	Commuting time: −0.025[c]; Commuting cost: 0.068[a];	−0.414[a]
Commuting time	−0.025[c]	Self-reported health: −0.010[a]; Working hours: −0.001[a]; Relationship spending: −0.001[b]; Relationship income: 0.006[a]; Social trust: −0.000;	−0.031[b]

a, b, and c denote significance at the 0.01, 0.05, and 0.1 levels, respectively.

modes does not change. This suggests that although the relationship between commuting time and cost and commuting mode can affect subjective well-being, this effect is limited. By controlling for travel time, Lades et al. (2020) found that the influence of short-distance travel modes (such as walking and biking) on subjective well-being decreases, while the effect of long-distance travel modes (such as car and public transit) on subjective well-being increases. That is, the difference between various commuting modes on subjective well-being is reduced when taking into account the complex relationship of commuting characteristics.

Researchers have identified the effect of commuting distance on mode choice and the effect of commuting mode on subjective well-being (Chng et al., 2016; Doering et al., 2019; Humagain and Singleton, 2020; Lades et al., 2020; Sun et al., 2022), but few studies have examined the effect of commuting distance on subjective well-being through commuting mode. We constructed such an impact path and confirmed the indirect effect of commuting distance on subjective well-being through commuting mode. Additionally, we found that commuting distance directly affects commuting time and commuting cost, thus affecting subjective well-being.

Taking the car mode as a reference, the indirect effect of choosing walking, bike, and e-bike is positive, while the indirect effect of choosing bus and metro is negative. This indicates that commuting distance affects how people choose to commute and produces different subjective well-being outcomes. Similar to our results, a study from Shanghai measured substitution effects between commuting modes and found that since active modes are commonly used for short commutes and bus are commonly used for long commutes, switching from bus to active modes significantly reduced respondents' subjective well-being due to the longer distance (Sun et al., 2022). Therefore, only focusing on the effect of commuting mode on subjective well-being may be biased, because it ignores the impact that the decision process of commuting mode (i.e., the effect of commuting distance on commuting mode) may have on subjective well-being.

We found that commuting time has an indirect impact on subjective well-being by influencing other life domains, which confirms the inter-domain transfer mechanism. The inter-domain transfer occurs when stress and discontent brought on by the commute spread to other life domains (Calderwood and Mitropoulos, 2021; Tao et al., 2022). The first life domain is health, which is measured by self-reported health. We found that longer commuting times reduce subjective well-being by affecting people's perceptions of health. Consistent with our results, Sun et al. (2021) found a significant mediating effect of self-reported health. Studies in the medical field have shown that employees with longer commuting times are more likely to report fatigue, decreased concentration, reduced sleep, stress, and mental health problems (Kageyama et al., 1998; Walsleben et al., 1999; Gottholmseder et al., 2009; Hansson et al., 2011; Han et al., 2023), and are also associated with objective physical health problems such as obesity and high blood pressure (Kunn-Nelen, 2016; De Vos et al., 2020). This suggests that the impact of commuting time spills over into the health domain, reducing the health status of residents and affecting subjective well-being. This result complements the findings of the critical review of the literature by Chatterjee et al. (2020) that there is no evidence that longer commuting times affect SWB through poorer physical health.

The second life domain is work, which is measured by working hours. We found a negative indirect effect of commuting time on subjective well-being through working hours, which is consistent with the findings of Nie and Sousa-Poza (2018). This suggests that the impact of long commuting times spills over into the work domain and then impacts subjective well-being. This conclusion can be explained by the resource drain model, which indicates that time and energy are finite resources that can only be dedicated to certain life domains (Delbosc, 2012). Long commuting times may cause inhabitants to expend more energy on commuting and work fewer hours, thus producing poor job performance and lowering subjective well-being (Calderwood and Mitropoulos, 2021). Additionally, long commuting times are associated with lower positive work emotions and worse post-work mood performance (Friman et al., 2017; Morris and Zhou, 2018).

The third life domain is community, which is measured by relationship spending, relationship income, and social trust. The direct effects of relationship spending and relationship income are significant, while the direct effect of social trust is insignificant. Relationship spending and relationship income represent one's position in social network relationships and the resources one can access. It can be seen as a form of social capital. Longer commuting time may lead to a decrease in the time of social participation, weakening social capital (Sun et al., 2021). This finding is consistent with previous research, which has found that longer commuting time can reduce social capital, leading to a decrease in subjective well-being (Delmelle et al., 2013; Mattisson et al., 2015).

6. Conclusion

This study proposes a theoretical framework for understanding the effect of commuting on subjective well-being and validates the mechanism and spillover effects of commuting on subjective well-being using the 2017 CHFS dataset. We found that the complex relationship between commuting characteristics has an impact on subjective well-being. The mediating effects of commuting mode, time, and cost narrow the gap between commuters' subjective well-being across different commuting modes. Commuting distance also affects commuting choices and produces varying subjective well-being outcomes. Furthermore, the impact of commuting spills over into health, work, and community domains, affecting subjective well-being.

Our findings have implications for urban planning and policy-making. We ranked commuters' subjective well-being using different commuting modes, with car, e-bike, walking, metro, bus, and bike showing varying levels of well-being from high to low. Accordingly, we proposed transportation interventions to improve the commuting experiences of different commuting modes. Enhancements to non-motorized infrastructure, such as rationalizing right-of-way and promoting independent bike lanes, can enhance the well-being of e-bike and bike commuters. For public transit commuters, measures like optimizing station locations, route organization, and scheduling can contribute to ensuring appropriate commuting times and a comfortable in-vehicle environment. Additionally, since commuting distance indirectly affects subjective well-being through its influence on commuting modes, urban planning should focus on developing accessible road networks and promoting high-density, compact urban forms to shorten residents' commuting distances. Moreover, as commuting time negatively affects subjective well-being, efforts should be made to reduce commuting times through initiatives like real-time transit information and access restrictions. Importantly, commuting time can also affect key life domains like health, work, and community, which are vital for public health, operational efficiency, and vibrancy. Government agencies should proactively address potential effects in these domains.

This study also has some limitations. First, the use of cross-sectional data prevents the establishment of a causal relationship between commuting and subjective well-being. To overcome this limitation, future research should consider longitudinal designs to examine causality. Second, due to data limitations, we did not consider the mediating role of commuting satisfaction, which reflects commuters' overall assessments of their commutes. Commuting satisfaction, as a domain-specific attitude, plays a crucial role in shaping commuters' perceptions and ultimately impacting their subjective well-being. These highlight avenues for future investigation. Exploring the mediating role of commuting satisfaction could provide further insights into the mechanisms underlying the relationship between commuting and subjective well-being. Third, also due to data limitation, we were unable to control for the effects of personality traits and psychological factors on the relationship between commuting and subjective well-being. Future research should consider personality traits and psychological factors to minimize potential bias caused by variable omission.

CRediT authorship contribution statement

Yiling Deng: Conceptualization, Data curation, Formal analysis, Writing – review & editing, Funding acquisition. **Liyang Wang:** Formal analysis, Writing – original draft. **Mengwei Chen:** Writing – review & editing, Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China [grant numbers 51978617].

References

Archer, M., Paleti, R., Konduri, K.C., Pendyala, R.M., Bhat, C.R., 2013. Modeling the connection between activity-travel patterns and subjective well-being. Transp. Res. Rec. J. Transp. Res. Board 2382, 102–111. https://doi.org/10.3141/2382-12.

Bai, C., Gong, Y., Feng, C., 2019. Social trust, pattern of difference, and subjective well-being. SAGE Open 9, 215824401986576. http://doi.org/10.1177/2158244019865765.

Calderwood, C., Mitropoulos, T., 2021. Commuting spillover: A systematic review and agenda for research. J. Organ. Behav. 42, 162–187. https://doi.org/10.1002/job.2462.

Chatterjee, K., Chng, S., Clark, B., Davis, A., De Vos, J., Ettema, D., Handy, S., Martin, A., Reardon, L., 2020. Commuting and wellbeing: A critical overview of the literature with implications for policy and future research. Transp. Rev. 40, 5–34. https://doi.org/10.1080/01441647.2019.1649317.

Chng, S., White, M., Abraham, C., Skippon, S., 2016. Commuting and wellbeing in London: The roles of commute mode and local public transport connectivity. Prev. Med. 88, 182–188. https://doi.org/10.1016/j.ypmed.2016.04.014.

Choi, J., Coughlin, J.F., D'Ambrosio, L., 2013. Travel time and subjective well-being. Transp. Res. Rec. 2537, 100–108. https://doi.org/10.3141/2357-12.

Clark, B., Chatterjee, K., Martin, A., Davis, A., 2020. How commuting affects subjective wellbeing. Transportation 47, 2777–2805. https://doi.org/10.1007/s11116-019-09983-9.

Clark, W.A.V., Yi, D., Huang, Y., 2019. Subjective well-being in China's changing society. Proc. Natl. Acad. Sci. 116, 16799–16804. https://doi.org/10.1073/pnas.1902926116.

Crespo, R.F., Mesurado, B., 2015. Happiness economics, eudaimonia and positive psychology: From happiness economics to flourishing economics. J. Happiness Stud. 16, 931–946. https://doi.org/10.1007/s10902-014-9541-4.

De Vos, J., Schwanen, T., Van Acker, V., Witlox, F., 2013. Travel and subjective well-being: A focus on findings, methods and future research needs. Transp. Rev. 33, 421–442. https://doi.org/10.1080/01441647.2013.815665.

De Vos, J., Singleton, P.A., Dill, J., 2020. Travel, health and well-being: A focus on past studies, a special issue, and future research. J. Transp. Health 19, 100973. https://doi.org/10.1016/j.jth.2020.100973.

Delbosc, A., 2012. The role of well-being in transport policy. Transp. Policy 23, 25–33. https://doi.org/10.1016/j.tranpol.2012.06.005.

Delmelle, E.C., Haslauer, E., Prinz, T., 2013. Social satisfaction, commuting and neighborhoods. J. Transp. Geogr. 30, 110–116. https://doi.org/10.1016/j.jtrangeo.2013.03.006.

Deng, Y., Li, W., Chen, M., 2023. Campus relocation as a natural experiment to investigate the determinants of commuting satisfaction. Travel Behav. Soc. 33, 100597 https://doi.org/10.1016/j.tbs.2023.100597.

Deng, Y., Zhao, P., 2022. Quantifying residential self-selection effects on commuting mode choice: A natural experiment. Transp. Res. Part D Transp. Environ. 104, 103197 https://doi.org/10.1016/j.trd.2022.103197.

Deng, Y., Zhao, P., 2023. The determinants of shared bike use in China. Transportation 50, 1–23. https://doi.org/10.1007/s11116-021-10235-y.

Di Tella, R., MacCulloch, R., 2006. Some uses of happiness data in economics. J. Econ. Perspect. 20, 25–46.

Dickerson, A., Hole, A.R., Munford, L.A., 2014. The relationship between well-being and commuting revisited: Does the choice of methodology matter? Reg. Sci. Urban Econ. 49, 321–329. https://doi.org/10.1016/j.regsciurbeco.2014.09.004.

Diener, E., 1984. Subjective well-being. Psychol. Bull. 95, 542–575. https://doi.org/10.1037/0033-2909.95.3.542.

Doering, L., Kroesen, M., Holz-Rau, C., 2019. The role of parents' mobility behavior for dynamics in car availability and commute mode use. Transportation 46, 957–994. https://doi.org/10.1007/s11116-017-9823-x.

Friman, M., Olsson, L.E., Ståhl, M., Ettema, D., Gärling, T., 2017. Travel and residual emotional well-being. Transp. Res. Part F Traffic Psychol. Behav. 49, 159–176. https://doi.org/10.1016/j.trf.2017.06.015.

Gao, Y., Rasouli, S., Timmermans, H., Wang, Y., 2017a. Understanding the relationship between travel satisfaction and subjective well-being considering the role of personality traits: A structural equation model. Transp. Res. Part F Traffic Psychol. Behav. 49, 110–123. https://doi.org/10.1016/j.trf.2017.06.005.

Gao, Y., Rasouli, S., Timmermans, H., Wang, Y., 2017b. Effects of traveller's mood and personality on ratings of satisfaction with daily trip stages. Travel Behav. Soc. 7, 1–11. https://doi.org/10.1016/j.tbs.2016.11.002.

Gao, Y., Rasouli, S., Timmermans, H., Wang, Y., 2022. A latent class structural equation model of the relationship between travel satisfaction and overall life satisfaction controlling for satisfaction with other life domains. Transportation. https://doi.org/10.1007/s11116-022-10324-6.

Gottholmseder, G., Nowotny, K., Pruckner, G.J., Theurl, E., 2009. Stress perception and commuting. Health Econ. 18, 559–576. https://doi.org/10.1002/hec.1389.

Green, A.E., 1997. A question of compromise? Case study evidence on the location and mobility strategies of dual career households. Reg. Stud. 31, 641–657. https://doi.org/10.1080/00343409750130731.

Han, L., Peng, C., Xu, Z., 2023. The effect of commuting time on quality of life: Evidence from China. Int. J. Environ. Res. Public. Health 20, 573. https://doi.org/10.3390/ijerph20010573.

Handy, S., Thigpen, C., 2019. Commute quality and its implications for commute satisfaction: Exploring the role of mode, location, and other factors. Travel Behav. Soc. 16, 241–248. https://doi.org/10.1016/j.tbs.2018.03.001.

Hansson, E., Mattisson, K., Bjork, J., Ostergren, P.-O., Jakobsson, K., 2011. Relationship between commuting and health outcomes in a cross-sectional population survey in southern Sweden. Bmc Public Health 11, 834. https://doi.org/10.1186/1471-2458-11-834.

Heinen, E., Maat, K., Van Wee, B., 2013. The effect of work-related factors on the bicycle commute mode choice in the Netherlands. Transportation 40, 23–43. https://doi.org/10.1007/s11116-012-9399-4.

Herman, K.M., Larouche, R., 2021. Active commuting to work or school: Associations with subjective well-being and work-life balance. J. Transp. Health 22, 101118. https://doi.org/10.1016/j.jth.2021.101118.

Hilbrecht, M., 2014. Highway to health? Commute time and wellbeing among Canadian adults. World Leis. J. 56, 151–163. https://doi.org/10.1080/16078055.2014.903723.

Humagain, P., Singleton, P.A., 2020. Investigating travel time satisfaction and actual versus ideal commute times: A path analysis approach. J. Transp. Health 16, 100829. https://doi.org/10.1016/j.jth.2020.100829.

Kageyama, T., Nishikido, N., Kobayashi, T., Kurokawa, Y., Kaneko, T., Kabuto, M., 1998. Long commuting time, extensive overtime, and sympathodominant state assessed in terms of short-term heart rate variability among male white-collar workers in the Tokyo megalopolis. Ind. Health 36, 209–217. https://doi.org/10.2486/indhealth.36.209.

Kahneman, D., Krueger, A.B., 2006. Developments in the measurement of subjective well-being. J. Econ. Perspect. 20, 3–24. https://doi.org/10.1257/089533006776526030.

Kahneman, D., Krueger, A.B., Schkade, D.A., Schwarz, N., Stone, A.A., 2004. A survey method for characterizing daily life experience: The day reconstruction method. Science 306, 1776–1780. https://doi.org/10.1126/science.1103572.
Kunn-Nelen, A., 2016. Does commuting affect health? Health Econ. 25, 984–1004. https://doi.org/10.1002/hec.3199.
Lades, L.K., Kelly, A., Kelleher, L., 2020. Why is active travel more satisfying than motorized travel? Evidence from Dublin. Transp. Res. Part A Policy Pract. 136, 318–333. https://doi.org/10.1016/j.tra.2020.04.007.
Le Vine, D.C., Wu, C., Polak, J., 2018. A nationwide study of factors associated with household car ownership in China. IATSS Res. 42, 128–137. https://doi.org/10.1016/j.iatssr.2017.10.001.
Liu, Y., Zhang, F., Wu, F., Liu, Y.e., Li, Z., 2017. The subjective wellbeing of migrants in Guangzhou, China: The impacts of the social and physical environment. Cities 60, 333–342. https://doi.org/10.1016/j.cities.2016.10.008.
Lorenz, O., 2018. Does commuting matter to subjective well-being? J. Transp. Geogr. 66, 180–199. https://doi.org/10.1016/j.jtrangeo.2017.11.019.
Ma, L., Zhang, X., Ding, X., Wang, G., 2018. Bike sharing and users' subjective well-being: An empirical study in China. Transp. Res. Part A Policy Pract. 118, 14–24. https://doi.org/10.1016/j.tra.2018.08.040.
Mao, Z., Ettema, D., Dijst, M., 2016. Commuting trip satisfaction in Beijing: Exploring the influence of multimodal behavior and modal flexibility. Transp. Res. Part A Policy Pract. 94, 592–603. https://doi.org/10.1016/j.tra.2016.10.017.
Mattisson, K., Hakansson, C., Jakobsson, K., 2015. Relationships between commuting and social capital among men and women in southern Sweden. Environ. Behav. 47, 734–753. https://doi.org/10.1177/0013916514529969.
Mattisson, K., Idris, A.O., Cromley, E., Håkansson, C., Östergren, P.-O., Jakobsson, K., 2018. Modelling the association between health indicators and commute mode choice: a cross-sectional study in southern Sweden. J. Transp. Health 11, 110–121. https://doi.org/10.1016/j.jth.2018.10.012.
Mîndrilă, D., 2010. Maximum Likelihood (ML) and diagonally weighted least squares (DWLS) estimation procedures: A comparison of estimation bias with ordinal and multivariate non-normal data. Int. J. Digit. Soc. 1, 60–66. https://doi.org/10.20533/ijds.2040.2570.2010.0010.
Morris, E.A., Zhou, Y., 2018. Are long commutes short on benefits? Commute duration and various manifestations of well-being. Travel Behav. Soc. 11, 101–110. https://doi.org/10.1016/j.tbs.2018.02.001.
Mouratidis, K., Ettema, D., Næss, P., 2019. Urban form, travel behavior, and travel satisfaction. Transp. Res. Part A Policy Pract. 129, 306–320. https://doi.org/10.1016/j.tra.2019.09.002.
Nie, P., Sousa-Poza, A., 2018. Commute time and subjective well-being in urban China. China Econ. Rev. 48, 188–204. https://doi.org/10.1016/j.chieco.2016.03.002.
Plaut, P.O., 2006. The intra-household choices regarding commuting and housing. Transp. Res. Part A Policy Pract. 40, 561–571. https://doi.org/10.1016/j.tra.2005.10.001.
Roberts, J., Hodgson, R., Dolan, P., 2011. "It's driving her mad": Gender differences in the effects of commuting on psychological health. J. Health Econ. 30, 1064–1076. https://doi.org/10.1016/j.jhealeco.2011.07.006.
Sandow, E., 2008. Commuting behaviour in sparsely populated areas: Evidence from northern Sweden. J. Transp. Geogr. 16, 14–27. https://doi.org/10.1016/j.jtrangeo.2007.04.004.
Sandow, E., Westin, K., 2010. The persevering commuter – Duration of long-distance commuting. Transp. Res. Part A Policy Pract. 44, 433–445. https://doi.org/10.1016/j.tra.2010.03.017.
Schwanen, T., Mokhtarian, P.L., 2005. What affects commute mode choice: neighborhood physical structure or preferences toward neighborhoods? J. Transp. Geogr. 13, 83–99. https://doi.org/10.1016/j.jtrangeo.2004.11.001.
Singleton, P.A., 2019. Walking (and cycling) to well-being: Modal and other determinants of subjective well-being during the commute. Travel Behav. Soc. 16, 249–261. https://doi.org/10.1016/j.tbs.2018.02.005.
Sirgy, M.J., 2001. Handbook of quality-of-life research: An ethical marketing perspective. Kluwer Academic Publishers, Dordrecht.
So, K.S., Orazem, P.F., Otto, D.M., 2001. The effects of housing prices, wages, and commuting time on joint residential and job location choices. Am. J. Agric. Econ. 83, 1036–1048. https://doi.org/10.1111/0002-9092.00228.
St-Louis, E., Manaugh, K., van Lierop, D., El-Geneidy, A., 2014. The happy commuter: A comparison of commuter satisfaction across modes. Transp. Res. Part F Traffic Psychol. Behav. 26, 160–170. https://doi.org/10.1016/j.trf.2014.07.004.
Stutzer, A., Frey, B.S., 2008. Stress that doesn't pay: The commuting paradox. Scand. J. Econ. 110, 339–366. https://doi.org/10.1111/j.1467-9442.2008.00542.x.
Sun, B., Lin, J., Yin, C., 2021. How does commute duration affect subjective well-being? A case study of Chinese cities. Transportation 48, 885–908. https://doi.org/10.1007/s11116-020-10082-3.
Sun, S., Yao, Y., Xu, L., He, X., Duan, Z., 2022. The use of E-moped increases commute satisfaction and subjective well-being: Evidence from Shanghai, China. Transp. Policy 117, 60–73. https://doi.org/10.1016/j.tranpol.2022.01.010.
Tao, Y., Petrović, A., van Ham, M., 2022. Commuting behaviours and subjective wellbeing: A critical review of longitudinal research. Transp. Rev. 1–23 https://doi.org/10.1080/01441647.2022.2145386.
Urry, J., 2002. Mobility and proximity. Sociol.- J. Br. Sociol. Assoc. 36, 255–274. https://doi.org/10.1177/0038038502036002002.
VanderWeele, T.J., 2017. On the promotion of human flourishing. Proc. Natl. Acad. Sci. U. S. A. 114, 8148–8156. https://doi.org/10.1073/pnas.1702996114.
Veenhoven, R., 2015. Informed pursuit of happiness: What we should know, do know and can get to know. J. Happiness Stud. 16, 1035–1071. https://doi.org/10.1007/s10902-014-9560-1.
Walsleben, J.A., Norman, R.G., Novak, R.D., O'Malley, E.B., Rapoport, D.M., Strohl, K.P., 1999. Sleep habits of Long Island rail road commuters. Sleep 22, 728–734. https://doi.org/10.1093/sleep/22.6.728.
Wang, X., Shao, C., Yin, C., Guan, L., 2020. Built environment, life events and commuting mode shift: Focus on gender differences. Transp. Res. Part D Transp. Environ. 88, 102598 https://doi.org/10.1016/j.trd.2020.102598.
Wang, X., Shao, C., Yin, C., Dong, C., 2021a. Exploring the effects of the built environment on commuting mode choice in neighborhoods near public transit stations: Evidence from China. Transp. Plan. Technol. 44, 111–127. https://doi.org/10.1080/03081060.2020.1851453.
Wang, X., Yin, C., Shao, C., 2021b. Relationships among haze pollution, commuting behavior and life satisfaction: A quasi-longitudinal analysis. Transp. Res. Part D Transp. Environ. 92, 102723 https://doi.org/10.1016/j.trd.2021.102723.
Wang, L., Zhang, S., Sun, W., Chen, C.-L., 2020. Exploring the physical and mental health of high-speed rail commuters: Suzhou-Shanghai inter-city commuting. J. Transp. Health 18, 100902. https://doi.org/10.1016/j.jth.2020.100902.
Wheatley, D., 2014. Travel-to-work and subjective well-being: A study of UK dual career households. J. Transp. Geogr. 39, 187–196. https://doi.org/10.1016/j.jtrangeo.2014.07.009.
Yang, W., Zhen, X., Zhou, S., 2021. How does neighborhood walkability affect obesity? The mediating role of commute mode. J. Transp. Land Use 14, 737–759. https://doi.org/10.5198/jtlu.2021.1948.
Ye, R., Titheridge, H., 2020. Impact of individuals' commuting trips on satisfaction and subjective well-being: Evidence from Xi'an, China. In: Handbook on Transport and Urban Transformation in China. pp. 266–285. https://doi.org/10.4337/9781786439246.00024.
Ye, R., De Vos, J., Ma, L., 2020. Analysing the association of dissonance between actual and ideal commute time and commute satisfaction. Transp. Res. Part A Policy Pract. 132, 47–60. https://doi.org/10.1016/j.tra.2019.10.011.
Ye, R., De Vos, J., Ma, L., 2022. New insights in travel satisfaction research. Transp. Res. Part D Transp. Environ. 102, 103163 https://doi.org/10.1016/j.trd.2021.103163.
Ye, R., Titheridge, H., 2017. Satisfaction with the commute: The role of travel mode choice, built environment and attitudes. Transp. Res. Part D Transp. Environ. 52, 535–547. https://doi.org/10.1016/j.trd.2016.06.011.
Yin, J., Cao, X. (Jason), Huang, X., 2021. Association between subway and life satisfaction: Evidence from Xi'an, China. Transp. Res. Part D Transp. Environ. 96, 102869. https://doi.org/10.1016/j.trd.2021.102869.
Yin, C., Shao, C., Dong, C., Wang, X., 2019. Happiness in urbanizing China: The role of commuting and multi-scale built environment across urban regions. Transp. Res. Part D Transp. Environ. 74, 306–317. https://doi.org/10.1016/j.trd.2019.08.010.

Zhang, C., Zhang, F., 2019. Effects of housing wealth on subjective well-being in urban China. J. Hous. Built Environ. 34, 965–985. https://doi.org/10.1007/s10901-019-09651-5.

Zheng, X., Yuan, Z., Zhang, X., 2020. Does happiness dwell in an owner-occupied house? Homeownership and subjective well-being in urban China. Cities 96, 102404. https://doi.org/10.1016/j.cities.2019.102404.

Zhu, J., Fan, Y., 2018. Commute happiness in Xi'an, China: Effects of commute mode, duration, and frequency. Travel Behav. Soc. 11, 43–51. https://doi.org/10.1016/j.tbs.2018.01.001.

Zhu, Z., Li, Z., Chen, H., Liu, Y., Zeng, J., 2019. Subjective well-being in China: How much does commuting matter? Transportation 46, 1505–1524. https://doi.org/10.1007/s11116-017-9848-1.

Zijlstra, T., Verhetsel, A., 2021. The commuters' burden: The relationship between commuting and wellbeing in Europe. Travel Behav. Soc. 23, 108–119. https://doi.org/10.1016/j.tbs.2020.12.007.